

procedure with the octonions, constructing seven orthonormal tangent vector fields to the unit sphere in $\mathbb{O}^m$ via an orthonormal basis $1, i, j, k, \dots$ for $\mathbb{O}$.

The upper bound of $2^r - 1$ for the number of orthonormal vector fields on S^{n-1} is not best possible in the remaining case $n \equiv 0 \pmod{16}$. The optimal upper bound is obtained instead using K-theory; see [VBKT] or [Husemoller 1966]. The construction of the requisite number of vector fields is again algebraic, this time using Clifford algebras.

Example 4L.6: A Map of mod p Hopf Invariant One. Let us describe a construction for a map $f: S^{2p} \rightarrow S^3$ such that in the mapping cone $C_f = S^3 \cup_f e^{2p+1}$, the first Steenrod power $P^1: H^3(C_f; \mathbb{Z}_p) \rightarrow H^{2p+1}(C_f; \mathbb{Z}_p)$ is nonzero, hence f is nonzero in π_{2p-3}^S . The construction starts with the fact that a generator of $H^2(K(\mathbb{Z}_p, 1); \mathbb{Z}_p)$ has nontrivial p^{th} power, so the operation $P^1: H^2(K(\mathbb{Z}_p, 1); \mathbb{Z}_p) \rightarrow H^{2p}(K(\mathbb{Z}_p, 1); \mathbb{Z}_p)$ is nontrivial by property (5). This remains true after we suspend to $\Sigma K(\mathbb{Z}_p, 1)$, and we showed in Proposition 4L.3 that $\Sigma K(\mathbb{Z}_p, 1)$ has the homotopy type of a wedge sum of CW complexes X_i , $1 \leq i \leq p-1$, with $\tilde{H}_*(X_i; \mathbb{Z})$ consisting only of a $\mathbb{Z}_p$ in each dimension congruent to $2i \pmod{2p-2}$. We are interested here in the space $X = X_1$, which has nontrivial $\mathbb{Z}_p$ cohomology in dimensions $2, 3, 2p, 2p+1, \dots$. Since X is, up to homotopy, a wedge summand of $\Sigma K(\mathbb{Z}_p, 1)$, the operation $P^1: H^3(X; \mathbb{Z}_p) \rightarrow H^{2p+1}(X; \mathbb{Z}_p)$ is nontrivial. Since X is simply-connected, the construction in §4.C shows that we may take X to have $(2p+1)$ -skeleton of the form $S^2 \cup e^3 \cup e^{2p} \cup e^{2p+1}$. In fact, using the notion of homology decomposition in §4.H, we can take this skeleton to be the reduced mapping cone C_g of a map of Moore spaces $g: M(\mathbb{Z}_p, 2p-1) \rightarrow M(\mathbb{Z}_p, 2)$. It follows that the quotient C_g/S^2 is the reduced mapping cone of the composition $h: M(\mathbb{Z}_p, 2p-1) \xrightarrow{g} M(\mathbb{Z}_p, 2) \rightarrow M(\mathbb{Z}_p, 2)/S^2 = S^3$. The restriction $h|_{S^{2p-1}}$ represents an element of $\pi_{2p-1}^S(S^3)$ that is either trivial or has order p since this restriction extends over the $2p$ -cell of $M(\mathbb{Z}_p, 2p-1)$ which is attached by a map $S^{2p-1} \rightarrow S^{2p-1}$ of degree p . In fact, $h|_{S^{2p-1}}$ is nullhomotopic since, as we will see in [SSAT] using the Serre spectral sequence, $\pi_i(S^3)$ contains no elements of order p if $i \leq 2p-1$. This implies that the space $C_h = C_g/S^2$ is homotopy equivalent to a CW complex Y obtained from $S^3 \vee S^{2p}$ by attaching a cell e^{2p+1} . The quotient Y/S^{2p} then has the form $S^3 \cup e^{2p+1}$, so it is the mapping cone of a map $f: S^{2p} \rightarrow S^3$. By construction there is a map $C_g \rightarrow C_f$ inducing an isomorphism on $\mathbb{Z}_p$ cohomology in dimensions 3 and $2p+1$, so the operation P^1 is nontrivial in $H^*(C_f; \mathbb{Z}_p)$ since this was true for C_g , the $(2p+1)$ -skeleton of X .

Example 4L.7: Moore Spaces. Let us use the operation Sq^2 to show that for $n \geq 2$, the identity map of $M(\mathbb{Z}_2, n)$ has order 4 in the group of basepoint-preserving homotopy classes of maps $M(\mathbb{Z}_2, n) \rightarrow M(\mathbb{Z}_2, n)$, with addition defined via the suspension structure on $M(\mathbb{Z}_2, n) = \Sigma M(\mathbb{Z}_2, n-1)$. According to Proposition 4H.2, this group is the middle term of a short exact sequence, the remaining terms of which contain only

elements of order 2. Hence if the identity map of $M(\mathbb{Z}_2, n)$ has order 4, this short exact sequence cannot split.

In view of the short exact sequence just referred to, it will suffice to show that twice the identity map of $M(\mathbb{Z}_2, n)$ is not nullhomotopic. If twice the identity were nullhomotopic, then the mapping cone C of this map would have the homotopy type of $M(\mathbb{Z}_2, n) \vee \Sigma M(\mathbb{Z}_2, n)$. This would force $Sq^2: H^n(C; \mathbb{Z}_2) \rightarrow H^{n+2}(C; \mathbb{Z}_2)$ to be trivial since the source and target groups would come from different wedge summands. However, we will now show that this Sq^2 operation is nontrivial. Twice the identity map of $M(\mathbb{Z}_2, n)$ can be regarded as the smash product of the degree 2 map $S^1 \rightarrow S^1$, $z \mapsto z^2$, with the identity map of $M(\mathbb{Z}_2, n-1)$. If we smash the cofibration sequence $S^1 \rightarrow S^1 \rightarrow \mathbb{R}P^2$ for this degree 2 map with $M(\mathbb{Z}_2, n-1)$ we get the cofiber sequence $M(\mathbb{Z}_2, n) \rightarrow M(\mathbb{Z}_2, n) \rightarrow C$, in view of the identity $(X/A) \wedge Y = (X \wedge Y)/(A \wedge Y)$. This means we can view C as $\mathbb{R}P^2 \wedge M(\mathbb{Z}_2, n-1)$. The Cartan formula translated to cross products gives $Sq^2(\alpha \times \beta) = Sq^0 \alpha \times Sq^2 \beta + Sq^1 \alpha \times Sq^1 \beta + Sq^2 \alpha \times Sq^0 \beta$. This holds for smash products as well as ordinary products, by naturality. Taking α to be a generator of $H^1(\mathbb{R}P^2; \mathbb{Z}_2)$ and β a generator of $H^{n-1}(M(\mathbb{Z}_2, n-1); \mathbb{Z}_2)$, we have $Sq^2 \alpha = 0 = Sq^2 \beta$, but $Sq^1 \alpha$ and $Sq^1 \beta$ are nonzero since Sq^1 is the Bockstein. By the Künneth formula, $Sq^1 \alpha \times Sq^1 \beta$ then generates $H^{n+2}(\mathbb{R}P^2 \wedge M(\mathbb{Z}_2, n-1); \mathbb{Z}_2)$ and we are done.

Adem Relations and the Steenrod Algebra

When Steenrod squares or powers are composed, the compositions satisfy certain relations, unfortunately rather complicated, known as **Adem relations**:

$$\begin{aligned} Sq^a Sq^b &= \sum_j \binom{b-j-1}{a-2j} Sq^{a+b-j} Sq^j & \text{if } a < 2b \\ P^a P^b &= \sum_j (-1)^{a+j} \binom{(p-1)(b-j)-1}{a-pj} P^{a+b-j} P^j & \text{if } a < pb \\ P^a \beta P^b &= \sum_j (-1)^{a+j} \binom{(p-1)(b-j)}{a-pj} \beta P^{a+b-j} P^j \\ &\quad - \sum_j (-1)^{a+j} \binom{(p-1)(b-j)-1}{a-pj-1} P^{a+b-j} \beta P^j & \text{if } a \leq pb \end{aligned}$$

By convention, the binomial coefficient $\binom{m}{n}$ is taken to be zero if m or n is negative or if $m < n$. Also $\binom{m}{0} = 1$ for $m \geq 0$.

For example, taking $a = 1$ in the Adem relation for the Steenrod squares we have $Sq^1 Sq^b = (b-1)Sq^{b+1}$, so $Sq^1 Sq^{2i} = Sq^{2i+1}$ and $Sq^1 Sq^{2i+1} = 0$. The relations $Sq^1 Sq^{2i} = Sq^{2i+1}$ and $Sq^1 = \beta$ explain the earlier comment that Sq^{2i} is the analog of P^i for $p = 2$.

The **Steenrod algebra** $\mathcal{A}_2$ is defined to be the algebra over $\mathbb{Z}_2$ that is the quotient of the algebra of polynomials in the noncommuting variables $Sq^1, Sq^2, \dots$ by the two-sided ideal generated by the Adem relations, that is, by the polynomials given by the differences between the left and right sides of the Adem relations. In similar fashion, $\mathcal{A}_p$ for odd p is defined to be the algebra over $\mathbb{Z}_p$ formed by polynomials in the noncommuting variables $\beta, P^1, P^2, \dots$ modulo the Adem relations and the relation

$\beta^2 = 0$. Thus for every space X , $H^*(X; \mathbb{Z}_p)$ is a module over $\mathcal{A}_p$, for all primes p . The Steenrod algebra is a graded algebra, the elements of degree k being those that map $H^n(X; \mathbb{Z}_p)$ to $H^{n+k}(X; \mathbb{Z}_p)$ for all n .

The next proposition implies that $\mathcal{A}_2$ is generated as an algebra by the elements Sq^{2^k} , while $\mathcal{A}_p$ for p odd is generated by β and the elements P^{p^k} .

Proposition 4L.8. *There is a relation $Sq^i = \sum_{0 < j < i} a_j Sq^{i-j} Sq^j$ with coefficients $a_j \in \mathbb{Z}_2$ whenever i is not a power of 2. Similarly, if i is not a power of p there is a relation $P^i = \sum_{0 < j < i} a_j P^{i-j} P^j$ with $a_j \in \mathbb{Z}_p$.*

Proof: The argument is the same for $p = 2$ and p odd, so we describe the latter case. The idea is to write i as the sum $a + b$ of integers $a > 0$ and $b > 0$ with $a < pb$, such that the coefficient of the $j = 0$ term in the Adem relation for $P^a P^b$ is nonzero. Then one can solve this relation for $P^{a+b} = P^i$.

Let the p -adic representation of i be $i = i_0 + i_1 p + \cdots + i_k p^k$ with $i_k \neq 0$. Let $b = p^k$ and $a = i - p^k$, so $b > 0$ and $a > 0$ if i is not a power of p . The claim is that $\binom{(p-1)b-1}{a}$ is nonzero in $\mathbb{Z}_p$. The p -adic expansion of $(p-1)b-1 = (p^{k+1}-1) - p^k$ is $(p-1) + (p-1)p + \cdots + (p-2)p^k$, and the p -adic expansion of a is $i_0 + i_1 p + \cdots + (i_k-1)p^k$. Hence $\binom{(p-1)b-1}{a} \equiv \binom{p-1}{i_0} \cdots \binom{p-2}{i_k-1}$ and in each factor of the latter product the numerator is nonzero in $\mathbb{Z}_p$ so the product is nonzero in $\mathbb{Z}_p$. When $p = 2$ the last factor is omitted, and the product is still nonzero in $\mathbb{Z}_2$. $\square$

This proposition says that most of the Sq^i 's and P^i 's are decomposable, where an element a of a graded algebra such as $\mathcal{A}_p$ is **decomposable** if it can be expressed in the form $\sum_i a_i b_i$ with each a_i and b_i having lower degree than a . The operation Sq^{2^k} is indecomposable since for α a generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2)$ we saw that $Sq^{2^k}(\alpha^{2^k}) = \alpha^{2^{k+1}}$ but $Sq^i(\alpha^{2^k}) = 0$ for $0 < i < 2^k$. Similarly P^{p^k} is indecomposable since if $\alpha \in H^2(\mathbb{C}P^\infty; \mathbb{Z}_p)$ is a generator then $P^{p^k}(\alpha^{p^k}) = \alpha^{p^{k+1}}$ but $P^i(\alpha^{p^k}) = 0$ for $0 < i < p^k$ and also $\beta(\alpha^{p^k}) = 0$.

Here is an application of the preceding proposition:

Theorem 4L.9. *Suppose $H^*(X; \mathbb{Z}_p)$ is the polynomial algebra $\mathbb{Z}_p[\alpha]$ on a generator α of dimension n , possibly truncated by the relation $\alpha^m = 0$ for $m > p$. Then if $p = 2$, n must be a power of 2, and if p is an odd prime, n must be of the form $p^k \ell$ where ℓ is an even divisor of $2(p-1)$.*

As we mentioned in §3.2, there is a stronger theorem that n must be 1, 2, 4, or 8 when $p = 2$, and n must be an even divisor of $2(p-1)$ when p is an odd prime. We also gave examples showing the necessity of the hypothesis $m > p$ in the case of a truncated polynomial algebra.

Proof: In the case $p = 2$, $Sq^n(\alpha) = \alpha^2 \neq 0$. If n is not a power of 2 then Sq^n decomposes into compositions $Sq^{n-j} Sq^j$ with $0 < j < n$. Such compositions must be zero since they pass through the group $H^{n+j}(X; \mathbb{Z}_2)$ which is zero for $0 < j < n$.

For odd p , the fact that α^2 is nonzero implies that n is even, say $n = 2k$. Then $P^k(\alpha) = \alpha^p \neq 0$. Since P^k can be expressed in terms of P^{p^i} 's, some P^{p^i} must be nonzero in $H^*(X; \mathbb{Z}_p)$. This implies that $2p^i(p-1)$, the amount by which P^{p^i} raises dimension, must be a multiple of n since $H^*(X; \mathbb{Z}_p)$ is concentrated in dimensions that are multiples of n . Since n divides $2p^i(p-1)$, it must be a power of p times a divisor of $2(p-1)$, and this divisor must be even since n is even and p is odd. $\square$

Corollary 4L.10. *If $H^*(X; \mathbb{Z})$ is a polynomial algebra $\mathbb{Z}[\alpha]$, possibly truncated by $\alpha^m = 0$ with $m > 3$, then $|\alpha| = 2$ or 4 .*

Proof: Passing from $\mathbb{Z}$ to $\mathbb{Z}_2$ coefficients, the theorem implies that $|\alpha|$ is a power of 2, and taking $\mathbb{Z}_3$ coefficients we see that $|\alpha|$ is a power of 3 times a divisor of $2(3-1) = 4$. $\square$

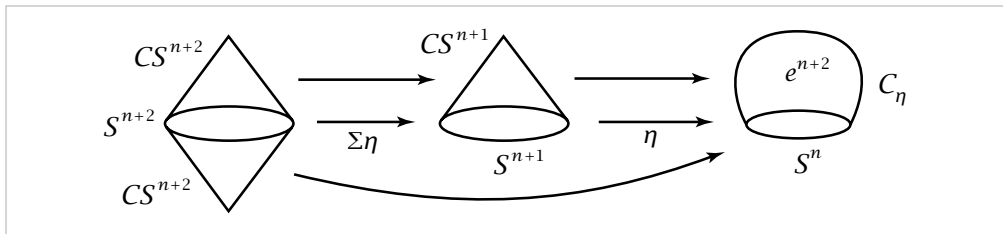
In particular, the octonionic projective plane $\mathbb{O}P^2$, constructed in Example 4.47 by attaching a 16-cell to S^8 via the Hopf map $S^{15} \rightarrow S^8$, does not generalize to an octonionic projective n -space $\mathbb{O}P^n$ with $n \geq 3$.

In a similar vein, decomposability implies that if an element of π_*^S is detected by a Sq^i or P^i then i must be a power of 2 for Sq^i and a power of p for P^i . For if Sq^i is decomposable, then the map $Sq^i: H^n(C_f; \mathbb{Z}_2) \rightarrow H^{n+i}(C_f; \mathbb{Z}_2)$ must be trivial since it is a sum of compositions that pass through trivial cohomology groups, and similarly for P^i .

Interestingly enough, the Adem relations can also be used in a positive way to detect elements of π_*^S , as the proof of the following result will show.

Proposition 4L.11. *If $\eta \in \pi_1^S$ is represented by the Hopf map $S^3 \rightarrow S^2$, then η^2 is nonzero in π_2^S . Similarly, the other two Hopf maps represent elements $\nu \in \pi_3^S$ and $\sigma \in \pi_7^S$ whose squares are nontrivial in π_6^S and π_{14}^S .*

Proof: Let $\eta: S^{n+1} \rightarrow S^n$ be a suspension of the Hopf map, with mapping cone C_η obtained from S^n by attaching a cell e^{n+2} via η . If we assume the composition $(\Sigma\eta)\eta$ is nullhomotopic, then we can define a map $f: S^{n+3} \rightarrow C_\eta$ in the following way. Decompose S^{n+3} as the union of two cones CS^{n+2} . On one of these cones let f be a nullhomotopy of $(\Sigma\eta)\eta$. On the other cone let f be the composition $CS^{n+2} \rightarrow CS^{n+1} \rightarrow C_\eta$ where the first map is obtained by coning $\Sigma\eta$ and the second map is a characteristic map for the cell e^{n+2} .



We use the map f to attach a cell e^{n+4} to C_η , forming a space X . This has C_η as its $(n+2)$ -skeleton, so $Sq^2: H^n(X; \mathbb{Z}_2) \rightarrow H^{n+2}(X; \mathbb{Z}_2)$ is an isomorphism. The map $Sq^2: H^{n+2}(X; \mathbb{Z}_2) \rightarrow H^{n+4}(X; \mathbb{Z}_2)$ is also an isomorphism since the quotient map $X \rightarrow X/S^n$ induces an isomorphism on cohomology groups above dimension n and X/S^n is homotopy equivalent to the mapping cone of $\Sigma^2 \eta$. Thus the composition $Sq^2 Sq^2: H^n(X; \mathbb{Z}_2) \rightarrow H^{n+4}(X; \mathbb{Z}_2)$ is an isomorphism. But this is impossible in view of the Adem relation $Sq^2 Sq^2 = Sq^3 Sq^1$, since Sq^1 is trivial on $H^n(X; \mathbb{Z}_2)$.

The same argument shows that ν^2 and σ^2 are nontrivial using the relations $Sq^4 Sq^4 = Sq^7 Sq^1 + Sq^6 Sq^2$ and $Sq^8 Sq^8 = Sq^{15} Sq^1 + Sq^{14} Sq^2 + Sq^{12} Sq^4$. $\square$

This line of reasoning does not work for odd primes and the element $\alpha \in \pi_{2p-3}^s$ detected by P^1 since the Adem relation for $P^1 P^1$ is $P^1 P^1 = 2P^2$, which is not helpful. And in fact $\alpha^2 = 0$ by the commutativity property of the product in π_*^s .

When dealing with $\mathcal{A}_2$ it is often convenient to abbreviate notation by writing a monomial $Sq^{i_1} Sq^{i_2} \cdots$ as Sq^I where I is the finite sequence of nonnegative integers $i_1, i_2, \dots$. Call Sq^I **admissible** if no Adem relation can be applied to it, that is, if $i_j \geq 2i_{j+1}$ for all j . The Adem relations imply that every monomial Sq^I can be written as a sum of admissible monomials. For if Sq^I is not admissible, it contains a pair $Sq^a Sq^b$ to which an Adem relation can be applied, yielding a sum of terms Sq^J for which $J > I$ with respect to the lexicographic ordering on finite sequences of integers. These Sq^J 's have the same degree $i_1 + \cdots + i_k$ as Sq^I , and since the number of monomials Sq^I of a fixed degree is finite, successive applications of the Adem relations eventually reduce any Sq^I to a sum of admissible monomials.

For odd p , elements of $\mathcal{A}_p$ are linear combinations of monomials $\beta^{\varepsilon_1} P^{i_1} \beta^{\varepsilon_2} P^{i_2} \cdots$ with each $\varepsilon_j = 0$ or 1 . Such a monomial is **admissible** if $i_j \geq \varepsilon_{j+1} + p i_{j+1}$ for all j , which again means that no Adem relation can be applied to the monomial. As with $\mathcal{A}_2$, the Adem relations suffice to reduce every monomial to a linear combination of admissible monomials, by the same argument as before but now using the lexicographic ordering on tuples $(\varepsilon_1 + p i_1, \varepsilon_2 + p i_2, \dots)$.

Define the **excess** of the admissible monomial Sq^I to be $\sum_j (i_j - 2i_{j+1})$, the amount by which Sq^I exceeds being admissible. For odd p one might expect the excess of an admissible monomial $\beta^{\varepsilon_1} P^{i_1} \beta^{\varepsilon_2} P^{i_2} \cdots$ to be defined as $\sum_j (i_j - p i_{j+1} - \varepsilon_{j+1})$, but instead it is defined to be $\sum_j (2i_j - 2p i_{j+1} - \varepsilon_{j+1})$, for reasons which will become clear below.

As we explained at the beginning of this section, cohomology operations correspond to elements in the cohomology of Eilenberg-MacLane spaces. Here is a rather important theorem which will be proved in [SSAT] since the proof makes heavy use of spectral sequences:

Theorem. For each prime p , $H^*(K(\mathbb{Z}_p, n); \mathbb{Z}_p)$ is the free commutative algebra on the generators $\Theta(\iota_n)$ where $\iota_n \in H^n(K(\mathbb{Z}_p, n); \mathbb{Z}_p)$ is a generator and Θ ranges over all admissible monomials of excess less than n .

Here ‘free commutative algebra’ means ‘polynomial algebra’ when $p = 2$ and ‘polynomial algebra on even-dimensional generators tensor exterior algebra on odd-dimensional generators’ when p is odd. We will say something about the rationale behind the ‘excess less than n ’ condition in a moment.

Specializing the theorem to the first two cases $n = 1, 2$, we have the following cohomology algebras:

$$\begin{aligned} K(\mathbb{Z}_2, 1) &: \mathbb{Z}_2[\iota] \\ K(\mathbb{Z}_p, 1) &: \Lambda_{\mathbb{Z}_p}[\iota] \otimes \mathbb{Z}_p[\beta\iota] \\ K(\mathbb{Z}_2, 2) &: \mathbb{Z}_2[\iota, Sq^1\iota, Sq^2Sq^1\iota, Sq^4Sq^2Sq^1\iota, \dots] \\ K(\mathbb{Z}_p, 2) &: \mathbb{Z}_p[\iota, \beta P^1\beta\iota, \beta P^p P^1\beta\iota, \beta P^{p^2} P^p P^1\beta\iota, \dots] \\ &\quad \otimes \Lambda_{\mathbb{Z}_p}[\beta\iota, P^1\beta\iota, P^p P^1\beta\iota, P^{p^2} P^p P^1\beta\iota, \dots] \end{aligned}$$

The theorem implies that the admissible monomials in $\mathcal{A}_p$ are linearly independent, hence form a basis for $\mathcal{A}_p$ as a vector space over $\mathbb{Z}_p$. For if some linear combination of admissible monomials were zero, then it would be zero when applied to the class ι_n , but if we choose n larger than the excess of each monomial in the linear combination, this would contradict the freeness of the algebra $H^*(K(\mathbb{Z}_p, n); \mathbb{Z}_p)$. Even though the multiplicative structure of the Steenrod algebra is rather complicated, the Adem relations provide a way of performing calculations algorithmically by systematically reducing all products to sums of admissible monomials. A proof of the linear independence of admissible monomials using more elementary techniques can be found in [Steenrod & Epstein 1962].

Another consequence of the theorem is that all cohomology operations with $\mathbb{Z}_p$ coefficients are polynomials in the Sq^i ’s when $p = 2$ and polynomials in the P^i ’s and β when p is odd, in view of Proposition 4L.1. We can also conclude that $\mathcal{A}_p$ consists precisely of all the $\mathbb{Z}_p$ cohomology operations that are stable, commuting with suspension. For consider the map $\Sigma K(\mathbb{Z}_p, n) \rightarrow K(\mathbb{Z}_p, n+1)$ that pulls ι_{n+1} back to the suspension of ι_n . This map induces an isomorphism on homotopy groups π_i for $i \leq 2n$ and a surjection for $i = 2n+1$ by Corollary 4.24, hence the same is true for homology and cohomology. Letting n go to infinity, the limit $\varinjlim \tilde{H}^*(K(\mathbb{Z}_p, n); \mathbb{Z}_p)$ then exists in a strong sense. On the one hand, this limit is exactly the stable operations by Proposition 4K.1 and the definition of a stable operation. On the other hand, the preceding theorem implies that this limit is $\mathcal{A}_p$ since it says that all elements of $H^*(K(\mathbb{Z}_p, n); \mathbb{Z}_p)$ below dimension $2n$ are uniquely expressible as sums of admissible monomials applied to ι_n .

Now let us explain why the condition ‘excess less than n ’ in the theorem is natural. For a monomial $Sq^I = Sq^{i_1} Sq^{i_2} \dots$ the definition of the excess $e(I)$ can be rewritten as

an equation $i_1 = e(I) + i_2 + i_3 + \cdots$. Thus if $e(I) > n$, we have $i_1 > |Sq^{i_2} Sq^{i_3} \cdots (\iota_n)|$, hence $Sq^I(\iota_n) = 0$. And if $e(I) = n$ then $Sq^I(\iota_n) = (Sq^{i_2} Sq^{i_3} \cdots (\iota_n))^2$ and either $Sq^{i_2} Sq^{i_3} \cdots$ has excess less than n or it has excess equal to n and we can repeat the process to write $Sq^{i_2} Sq^{i_3} \cdots (\iota_n) = (Sq^{i_3} \cdots (\iota_n))^2$, and so on, until we obtain an equation $Sq^I(\iota_n) = (Sq^J(\iota_n))^{2^k}$ with $e(J) < n$, so that $Sq^I(\iota_n)$ is already in the algebra generated by the elements $Sq^J(\iota_n)$ with $e(J) < n$. The situation for odd p is similar. For an admissible monomial $P^I = \beta^{\varepsilon_1} P^{i_1} \beta^{\varepsilon_2} P^{i_2} \cdots$ the definition of excess gives $2i_1 = e(I) + \varepsilon_2 + 2(p-1)i_2 + \cdots$, so if $e(I) > n$ we must have $P^I(\iota_n) = 0$, and if $e(I) = n$ then either $P^I(\iota_n)$ is a power $(P^J(\iota_n))^{p^k}$ with $e(J) < n$, or, if P^I begins with β , then $P^I(\iota_n) = \beta((P^J(\iota_n))^{p^k}) = 0$ by the formula $\beta(x^m) = mx^{m-1}\beta(x)$, which is valid when $|x|$ is even, as we may assume is the case here, otherwise $(P^J(\iota_n))^{p^k} = 0$ by commutativity of cup product.

There is another set of relations among Steenrod squares equivalent to the Adem relations and somewhat easier to remember:

$$\sum_j \binom{k}{j} Sq^{2n-k-j-1} Sq^{n+j} = 0$$

When $k = 0$ this is simply the relation $Sq^{2n-1} Sq^n = 0$, and the cases $k > 0$ are obtained from this via Pascal's triangle. For example, from $Sq^7 Sq^4 = 0$ we obtain the following table of relations:

$$\begin{array}{rcl}
 Sq^7 Sq^4 & & = 0 \\
 Sq^6 Sq^4 + Sq^7 Sq^3 & & = 0 \\
 Sq^5 Sq^4 + Sq^7 Sq^2 & & = 0 \\
 Sq^4 Sq^4 + Sq^5 Sq^3 + Sq^6 Sq^2 + Sq^7 Sq^1 & & = 0 \\
 Sq^3 Sq^4 + Sq^7 Sq^0 & & = 0 \\
 Sq^2 Sq^4 + Sq^3 Sq^3 + Sq^6 Sq^0 & & = 0 \\
 Sq^1 Sq^4 + Sq^3 Sq^2 + Sq^5 Sq^0 & & = 0 \\
 Sq^0 Sq^4 + Sq^1 Sq^3 + Sq^2 Sq^2 + Sq^3 Sq^1 + Sq^4 Sq^0 & & = 0
 \end{array}$$

These relations are not in simplest possible form. For example, $Sq^5 Sq^3 = 0$ in the fourth row and $Sq^3 Sq^2 = 0$ in the seventh row, instances of $Sq^{2n-1} Sq^n = 0$. For Steenrod powers there are similar relations $\sum_j \binom{k}{j} P^{pn-k-j-1} P^{n+j} = 0$ derived from the basic relation $P^{pn-1} P^n = 0$. We leave it to the interested reader to show that these relations follow from the Adem relations.

Constructing the Squares and Powers

Now we turn to the construction of the Steenrod squares and powers, and the proof of their basic properties including the Adem relations. As will be seen, this all hinges on the fact that cohomology is maps into Eilenberg-MacLane spaces. The case $p = 2$ is in some ways simpler than the case p odd, so in the first part of the development we will specialize p to 2 whenever there is a significant difference between the two cases.

Before giving the detailed construction, let us describe the idea in the simplest case $p = 2$. The cup product square α^2 of an element $\alpha \in H^n(X; \mathbb{Z}_2)$ can be viewed as a composition $X \rightarrow X \times X \rightarrow K(\mathbb{Z}_2, 2n)$, with the first map the diagonal map and the second map representing the cross product $\alpha \times \alpha$. Since we have $\mathbb{Z}_2$ coefficients, cup product and cross product are strictly commutative, so if $T: X \times X \rightarrow X \times X$ is the map transposing the two factors, $T(x_1, x_2) = (x_2, x_1)$, then $T^*(\alpha \times \alpha) = \alpha \times \alpha$. Thinking of $\alpha \times \alpha$ as a map $X \times X \rightarrow K(\mathbb{Z}_2, 2n)$, this says there is a homotopy f_t from $\alpha \times \alpha$ to $(\alpha \times \alpha)T$. If we follow the homotopy f_t by the homotopy $f_t T$, we obtain a homotopy from $\alpha \times \alpha$ to $(\alpha \times \alpha)T$ and then to $(\alpha \times \alpha)T^2 = \alpha \times \alpha$, that is, a loop of maps $X \times X \rightarrow K(\mathbb{Z}_2, 2n)$. We can view this loop as a map $S^1 \times X \times X \rightarrow K(\mathbb{Z}_2, 2n)$. If the homotopy f_t is chosen appropriately, the loop of maps will be nullhomotopic, extending to a map $D^2 \times X \times X \rightarrow K(\mathbb{Z}_2, 2n)$. Regarding D^2 as the upper hemisphere of S^2 , this gives half of a map $S^2 \times X \times X \rightarrow K(\mathbb{Z}_2, 2n)$, and once again we obtain the other half by composition with T . With a little care, the process can be repeated infinitely often to yield a map $S^\infty \times X \times X \rightarrow K(\mathbb{Z}_2, 2n)$ with the property that each pair of points (s, x_1, x_2) and $(-s, x_2, x_1)$ is sent to the same point in $K(\mathbb{Z}_2, 2n)$. This means that when we compose with the evident map $S^\infty \times X \rightarrow S^\infty \times X \times X$, there is an induced quotient map $\mathbb{R}P^\infty \times X \rightarrow K(\mathbb{Z}_2, 2n)$. This map represents a class in $H^{2n}(\mathbb{R}P^\infty \times X; \mathbb{Z}_2)$. By the Künneth formula and the fact that $H^*(\mathbb{R}P^\infty; \mathbb{Z}_2)$ is the polynomial ring $\mathbb{Z}_2[\omega]$, this cohomology class in $H^{2n}(\mathbb{R}P^\infty \times X; \mathbb{Z}_2)$ can be written in the form $\sum_i \omega^{n-i} \times a_i$ with $a_i \in H^{n+i}(X; \mathbb{Z}_2)$. Then we define $Sq^i(\alpha) = a_i$.

The construction of the map $S^\infty \times X \times X \rightarrow K(\mathbb{Z}_2, 2n)$ will proceed cell by cell, so it will be convenient to eliminate any unnecessary cells. This is done by replacing $X \times X$ by the smash product $X \wedge X$ and factoring out a cross-sectional slice S^∞ in $S^\infty \times X \wedge X$. A further simplification will be to use naturality to reduce to the case $X = K(\mathbb{Z}_2, n)$.

Now we begin the actual construction. For a space X with basepoint x_0 , let $X^{\wedge p}$ denote the smash product $X \wedge \cdots \wedge X$ of p copies of X . There is a map $T: X^{\wedge p} \rightarrow X^{\wedge p}$, $T(x_1, \dots, x_p) = (x_2, \dots, x_p, x_1)$, permuting the factors cyclically. Note that when $p = 2$ this is just the transposition $(x_1, x_2) \mapsto (x_2, x_1)$. The map T generates an action of $\mathbb{Z}_p$ on $X^{\wedge p}$. There is also the standard action of $\mathbb{Z}_p$ on S^∞ viewed as the union of the unit spheres S^{2n-1} in $\mathbb{C}^n$, a generator of $\mathbb{Z}_p$ rotating each $\mathbb{C}$ factor through an angle $2\pi/p$, with quotient space an infinite-dimensional lens space L^∞ , or $\mathbb{R}P^\infty$ when $p = 2$. On the product $S^\infty \times X^{\wedge p}$ there is then the diagonal action $g(s, x) = (g(s), g(x))$ for $g \in \mathbb{Z}_p$. Let ΓX denote the orbit space $(S^\infty \times X^{\wedge p})/\mathbb{Z}_p$ of this diagonal action. This is the same as the Borel construction $S^\infty \times_{\mathbb{Z}_p} X^{\wedge p}$ described in §3.G. The projection $S^\infty \times X^{\wedge p} \rightarrow S^\infty$ induces a projection $\pi: \Gamma X \rightarrow L^\infty$ with $\pi^{-1}(z) = X^{\wedge p}$ for all $z \in L^\infty$ since the action of $\mathbb{Z}_p$ on S^∞ is free. This projection $\Gamma X \rightarrow L^\infty$ is in fact a fiber bundle, though we shall not need this fact and so we leave the proof as an exercise. The $\mathbb{Z}_p$ action on $X^{\wedge p}$ fixes the basepoint $x_0 \in X^{\wedge p}$, so the inclusion $S^\infty \times \{x_0\} \hookrightarrow S^\infty \times X^{\wedge p}$ induces an inclusion $L^\infty \hookrightarrow \Gamma X$. The composition $L^\infty \hookrightarrow \Gamma X \rightarrow L^\infty$ is the identity, so in